CHAPTER 12

LIMITS AND DERIVATIVES

CHAPTER ESSENCE AT A GLANCE:
Calculus is the mathematics of change! Limits let us inspect what happens as we get infinitely close to a point without actually touching it. The derivative f'(x) is simply the slope of the tangent line to the curve at that exact point!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Fundamental Trig & Algebraic Limits:** lim {x -> 0} [sin x over x] = 1 (x in radians!) ; lim {x -> a} [(x^n - a^n) over (x - a)] = n a^{n-1} ; lim {x -> 0} [(1 - cos x) over x^2] = 1 over 2.

• **First Principle of Derivative:** f'(x) = lim {h -> 0} [f(x + h) - f(x)] over h. Always substitute f(x+h), expand, cancel h from numerator and denominator, then set h = 0.

• **Power Rule of Derivatives:** d by dx [x^n] = n x^{n-1}. Special: d by dx(x) = 1, d by dx(c) = 0, d by dx(sqrt(x)) = 1 over (2sqrt(x)).

• **Product Rule:** d by dx [u · v] = u' v + u v' (Derivative of first × second + first × derivative of second).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\lim_{x\to 0}\frac{\sin x}{x}=1,\quad \lim_{x\to 0}\frac{1-\cos x}{x}=0\quad (x \text{ in radians})$$

Fundamental Trigonometric Limit Theorems

$$\lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}\quad \text{(Standard Algebraic Limit)}$$

Standard Polynomial Limit Identity

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}\quad \text{(Derivative from First Principles)}$$

Formal Definition of Derivative from First Principles

$$(uv)'=u'v+uv',\quad \left(\frac{u}{v}\right)'=\frac{u'v-uv'}{v^2}$$

Product Rule and Quotient Rule of Differential Calculus

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In the quotient rule, the numerator has a MINUS sign: (u' v - u v'). Also, in trig limits, x MUST be in radians!

• **Alert:** First Principle definition is a guaranteed 3-4 mark board derivation: f'(x) = lim(h -> 0) [f(x + h) - f(x)] over h!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find the derivative of f(x) = x^2 - 2 at x = 10. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Differentiation]** f'(x) = d by dx (x^2 - 2) = 2x - 0 = 2x. [1 Mark]

• **[Step 2: Value at x = 10]** f'(10) = 2(10) = 20. [1 Mark]

• **[Boxed Result: f'(10) = 20]**

2. Evaluate the limit: lim {x → 2} [($3x^2$ - x - 10) over (x^2 - 4)]. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Factoring 0 over 0 Form]** At x = 2, numerator = 3(4) - 2 - 10 = 0, denominator = 4 - 4 = 0 (0 over 0 indeterminate form). Numerator = $3x^2$ - 6x + 5x - 10 = 3x(x - 2) + 5(x - 2) = (x - 2)(3x + 5). Denominator = (x - 2)(x + 2). [1 Mark]

• **[Step 2: Cancel (x - 2) & Evaluate]** lim {x → 2} [(x - 2)(3x + 5)] over [(x - 2)(x + 2)] = lim {x → 2} [(3x + 5) over (x + 2)] = [3(2) + 5] over (2 + 2) = 11 over 4. [1 Mark]

• **[Boxed Result: 11 over 4]**

3. Find the derivative of f(x) = sin x from first principles. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: First Principle Definition]** f'(x) = lim {h → 0} [f(x + h) - f(x)] over h = lim {h → 0} [sin(x + h) - sin x] over h. [1 Mark]

• **[Step 2: Transformation Formula]** sin C - sin D = 2 cos((C+D) over 2) sin((C-D) over 2). sin(x + h) - sin x = 2 cos(x + h over 2) sin(h over 2). f'(x) = lim {h → 0} [2 cos(x + h over 2) sin(h over 2)] over h = lim {h → 0} cos(x + h over 2) x lim {h → 0} [sin(h over 2) over (h over 2)]. [1.5 Marks]

• **[Step 3: Evaluation]** = cos(x + 0) x 1 = cos x. [0.5 Mark]

• **[Boxed Result: d by dx (sin x) = cos x (from first principles)]**

4. Find the derivative of f(x) = cos x from first principles. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Definition Setup]** f'(x) = lim {h → 0} [cos(x + h) - cos x] over h. [1 Mark]

• **[Step 2: Transformation Formula]** cos C - cos D = -2 sin((C+D) over 2) sin((C-D) over 2). cos(x + h) - cos x = -2 sin(x + h over 2) sin(h over 2). f'(x) = lim {h → 0} [-2 sin(x + h over 2) sin(h over 2)] over h = - lim {h → 0} sin(x + h over 2) x lim {h → 0} [sin(h over 2) over (h over 2)]. [1.5 Marks]

• **[Step 3: Result]** = -sin(x) x 1 = -sin x. [0.5 Mark]

• **[Boxed Result: d by dx (cos x) = -sin x (from first principles)]**

5. Find the derivative of f(x) = x^n from first principles, where n is a positive integer. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Binomial Setup]** f'(x) = lim {h → 0} [$(x + h)^n$ - x^n] over h. [1 Mark]

• **[Step 2: Binomial Expansion]** $(x + h)^n$ = x^n + nC1 x^(n-1) h + nC2 x^(n-2) h^2 + ... + h^n. $(x + h)^n$ - x^n = n x^(n-1) h + h^2 [nC2 x^(n-2) + ...]. [1 Mark]

• **[Step 3: Dividing by h and Taking Limit]** f'(x) = lim {h → 0} [n x^(n-1) + h (nC2 x^(n-2) + ...)] = n x^(n-1) + 0 = n x^(n-1). [1 Mark]

• **[Boxed Result: d by dx (x^n) = n x^(n-1)]**

6. Find the derivative of f(x) = tan x from first principles. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Definition Setup]** f'(x) = lim {h → 0} [tan(x + h) - tan x] over h = lim {h → 0} [sin(x + h) over cos(x + h) - sin x over cos x] over h. [1 Mark]

• **[Step 2: LCM & Numerator Identity]** Numerator = sin(x + h) cos x - cos(x + h) sin x = sin[(x + h) - x] = sin h. [1.5 Marks]

• **[Step 3: Limit Evaluation]** f'(x) = lim {h → 0} [sin h over h] x [1 over (cos(x + h) cos x)] = 1 x [1 over (cos x cos x)] = 1 over $\cos^2$ x = $\sec^2$ x. [1.5 Marks]

• **[Boxed Result: d by dx (tan x) = $\sec^2$ x (from first principles)]**

7. Evaluate: (a) lim {x → 0} [sin(ax) over sin(bx)], a, b ≠ 0. (2 Marks)n(b) lim {x → 0} [(cos 2x - 1) over (cos x - 1)]. (2 Marks) **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Part (a) Evaluation]** lim {x → 0} [sin(ax) over sin(bx)] = lim {x → 0} [{sin(ax) over (ax)} x (ax)] over [{sin(bx) over (bx)} x (bx)] = (1 x a x) over (1 x b x) = a over b. [2 Marks]

• **[Step 2: Part (b) Evaluation]** 1 - cos 2x = 2 $\sin^2$ x, 1 - cos x = 2 $\sin^2$(x over 2). Expression = [-(1 - cos 2x)] over [-(1 - cos x)] = [2 $\sin^2$ x] over [2 $\sin^2$(x over 2)] = [$\sin^2$ x over x^2] over [$\sin^2$(x over 2) over x^2] = 1 over [(1 over 4) (sin(x over 2) over (x over 2))^2] = 1 over (1 over 4) = 4. [2 Marks]

• **[Boxed Result: (a) a over b, (b) 4]**